

Common Themes and Challenges

The vast majority of these works aim to personalize medication recommendation from EHRs while ensuring safety (the absence of harmful drug-drug interactions) and explainability. They use longitudinal data (MIMIC-III/IV in almost all cases) and aim to model the complex relationships among patients, diagnoses, procedures, and medications. They address recurring problems:

- **Sparse and rare data:** the need to use external sources or hierarchical structures (KGs, ontologies) to compensate for infrequent clinical codes [4, L61-L66] [29, L13-L21].
- **DDI safety:** most explicitly integrate drug-drug interaction graphs (GAMENet [14, L55-L63], SafeDrug [12, L57-L64], KEHGCN [6, L27-L30], HypeMed [8, L60-L68], EGNet [41, L52-L60]) or loss terms to avoid dangerous combinations.
- **Time-series/history models:** they model patient history either with RNN/GRU architectures (KGDNet, KindMed [38, L59-L67], MGRN [23, L169-L177]) or with Transformers, in order to exploit multi-visit clinical evolution.
- **Integration of medical knowledge:** many studies enrich EHR data with knowledge graphs or external ontologies (GraphCare [29, L13-L21], KindMed [38, L59-L67], HiRef [4, L61-L66], KEHGCN [6, L17-L25]). Some even use LLMs to extract knowledge (GraphCare) or pre-train on subgraphs (COMPASS).
- **Explainability:** several approaches provide interpretation components. For example, PIE-Med follows a "Predict -> Interpret -> Explain" pipeline, separating the recommendation phase (GNN) from the explanation phase (feature attribution with IG/GNNExplainer, then narrative justification through an LLM) [36, L133-L142] [36, L145-L154]. Other methods produce explanations based on similar neighbors or metapaths in the KG (EGNet cites similar patients [41, L52-L60], while KEHGCN exposes explanatory metapaths [6, L17-L25]).

Methodologies and Architectures

- **Personalized knowledge graphs:** GraphCare generates a personalized KG for each patient from external sources via an LLM, then uses a dual-attention GNN (BAT) [29, L13-L21]. KindMed merges external medical KGs and co-occurrence graphs, separately learning the clinical pathway and the medication pathway over time [38, L59-L67]. HiRef encodes hierarchical ontologies in hyperbolic space and cleans the co-occurrence graph with sparse regularization [4, L61-L70]. KEHGCN constructs a hierarchical hypergraph and uses selected metapaths from the KG to explain recommendations [6, L17-L25].
- **Convolution and hypergraphs:** GAMENet uses a graph-convolutional memory network to integrate the DDI KG as external memory [14, L55-L63]. KEHGCN and HypeMed use hypergraphs to capture multi-entity relationships within a visit [6, L17-L25] [8, L60-L68]. HypeMed adopts a two-step approach (contrastive pre-training on visits in MedRep, followed by dynamic retrieval of similar cases in SimMR) to strengthen intra-visit consistency and exploit previous visits [8, L60-L68].
- **Separate temporal flows:** KindMed separately models clinical evolution (diagnoses/procedures) and medication evolution through two RNNs before fusion [38, L59-L67]. MGRN constructs several "views" (visit-level, sequence-level, and token-level) to better represent patients and medications, and uses a gating-based retrieval module over past visits [23, L169-L177].
- **Attention mechanisms and LLMs:** some models use attention to aggregate patient history (KindMed [38, L59-L67], HypeMed, GraphCare). PIE-Med and COMPASS use LLMs: PIE-Med for LLM-based justification after attribution, while COMPASS (not listed here) pre-trains an LLM with text generated from graphs and then instruction-tunes it to extract user preferences. LEADER uses an LLM to suggest medications through prompts, then distills its knowledge into a more compact model [25, L49-L58] [25, L63-L71].
- **Objectives and outputs:** all aim to produce a set of medications (multi-label output). Training losses typically combine recommendation accuracy with a term penalizing undesirable DDIs (GAMENet,

SafeDrug). Systems evaluate Jaccard, AUC, F1-score, PR-AUC [23, L177-L180] [29, L17-L21], as well as the generated DDI rate (for example, GAMENet reports a 3.6% DDI reduction) [14, L59-L63].

Key Contributions and Differences

Each approach introduces specific innovations:

- **PIE-Med (2025)** clearly separates the prediction phase (a GNN on a heterogeneous graph through MIMIC) from the explanation phase. It uses Integrated Gradients and GNNExplainer for interpretation (node importance), plus a group of LLM-simulated experts to formulate the final explanation [36, L133-L142] [36, L145-L154]. This ensures an explanation grounded in the actual prediction rather than simple LLM generation.
- **GraphCare (ICLR 2024)** illustrates the use of personalized KGs extracted by LLM for clinical prediction. Its major gains are in predictive performance (for example, +10.8% F1 in medication recommendation on MIMIC-III) [29, L17-L21]. However, the explanation remains integrated into the GNN model (with no explicit clinician-facing interface), resulting in attention weights over clinical facts.
- **KindMed (2023)** combines external knowledge with dual hierarchical modeling (clinical vs. medication). Its novelties include the combination of external relational graphs and parallel temporal sequences [38, L59-L67], as well as an attention-based "attentive prescribing" scheme over three representation vectors: history, progression, and current state.
- **HiRef (2025, arXiv)** introduces hyperbolic-space embeddings for medical ontologies and refines the EHR co-occurrence graph through sparse regularization [4, L61-L70]. This approach improves robustness to rare medical codes or codes absent during out-of-distribution testing.
- **SafeDrug (IJCAI 2021)** focuses on recommendation safety by explicitly encoding the molecular structure of medications (molecular graphs) and controlling DDIs through a dedicated loss [12, L57-L64]. This significantly reduces negative interactions (-19.4% DDI) while maintaining quality.
- **GAMENet (AAAI 2019)** is one of the pioneers in integrating a DDI KG into recommendation through a GNN memory module [14, L55-L63]. Although older, it established the foundations for safety-oriented approaches.
- **KEHGCN (AAAI 2026)** constructs a hierarchical multi-level hypergraph to capture dependencies within EHRs, enriched by informative metapaths extracted from an external KG [6, L17-L25]. It is notable for explicitly considering "negative relations" (prohibited relations) in the KG in order to improve safety [6, L27-L30].
- **HypeMed (2026)** proposes a two-stage hypergraph framework: first, contrastive pre-training (MedRep), where each visit is encoded as a hyperedge; then, a retrieval phase (SimMR), which dynamically integrates similar previous visits at prediction time [8, L60-L68]. The idea is to balance global model learning with local search in the embedding space.
- **CausalMed (CIKM 2024)** infuses causality by discovering cause-and-effect links between diagnoses and medications, focusing recommendations on the patient's overall health state [10, L64-L72]. It models how each medication specifically treats a particular diagnosis in the patient's current state.
- **MGRN (Bioinformatics 2024)** uses a multi-view approach (visit/sequence/word) to enrich the patient-medication representation, and includes a gating retrieval module to extract key elements from past medication histories [23, L169-L177].
- **LEADER (2024)** relies on the semantic power of LLMs for recommendation. It creates prompts that allow the LLM to suggest medications, corrects the medication-specific out-of-vocabulary problem, and then distills the LLM into a lightweight model through feature-level distillation [25, L49-L58] [25, L63-L71].

Each method offers original solutions but also has limitations: few use explainability with clinical rigor (explanations often remain internal or vague), and none explicitly addresses contradictions between sources of evidence. Similarly, most are evaluated only on MIMIC (with little evaluation on eICU or other external databases), and they sometimes lack human validation.

Key Techniques and Concepts to Remember

- **Knowledge graphs and ontologies** (GraphCare [29, L13-L21], KindMed [38, L59-L67], HiRef [4, L61-L70]): integrating a UMLS or SNOMED KG improves coverage and generalization. Using semantic relationships (e.g., hierarchical ones) in hyperbolic space [4, L61-L66] helps manage unseen codes more effectively.
- **Hypergraphs and hierarchical views** (KEHGCN, HypeMed, MGRN): instead of simple pairwise edges, modeling multi-entity interactions per visit (hypergraphs of diagnoses/medications within the same encounter) and at multiple levels (global vs. local) strengthens coherence.
- **Safety through DDIs**: apply a loss term or a special module to minimize DDIs (SafeDrug [12, L57-L64], GAMENet [14, L55-L63], KEHGCN [6, L27-L30], EGNet [41, L52-L60]). Explicitly integrating negative "prohibition" edges in the KG (KEHGCN) is a good practice.
- **Dual patient/medication modeling**: separating the clinical trajectory and the medication trajectory (KindMed [38, L59-L67], MGRN) allows their respective dynamics to be captured before fusion.
- **Unsupervised and contrastive learning**: HypeMed pre-trains contrastively to structure the visit space. This type of pre-training can stabilize the representation before the recommendation task.
- **Multi-level explainability**: combine local attributes (LIME [43, L59-L63] on patient features, GNNExplainer [36, L141-L144] on graph structures) with evidence paths in the KG or literature (for example, extracting similar neighbors as EGNet does) and formal clinical rules (not covered in these works, but necessary for auditability).
- **Recommendation/explanation separation (PIE-Med)**: maintaining two separate pipelines ensures that the explanation is not merely a "made-up story" but first derives from the raw prediction. PIE's expert agents show the value of human-LLM review; even if your system aims for autonomy, the idea of validating and comparing different sources of explanation remains valuable [36, L145-L154].
- **LLM-based summarization**: using the LLM at the final stage to formulate explanations in natural language ("grounded summarization") can improve clinician acceptance. COMPASS and LEADER show how to align an LLM with a structured format (through prompts or distillation) [25, L49-L58]. However, generation outside the context must be limited, and the source of information must always be provided (e.g., bibliographic provenance or similarity metrics).
- **Rigorous evaluation**: all these studies rely on rigorous patient splits and avoid temporal leakage. Your planned setup (patient split, temporal cutoff, and feature-leakage-aware design) aligns with this approach. It is crucial to measure not only accuracy (F1, AUC, PRAUC), but also clinical relevance (e.g., DDI, guideline coverage, necessary warnings), and to include validation on an external dataset (eICU).

Concrete Recommendations for Your System

To address the research question of accuracy, faithfulness, grounding, and clinical intelligibility, integrate the following ideas:

- **Explicit medical knowledge**: build a patient-specific KG (as in GraphCare) by enriching dialogue/EHR data through existing databases. For example, query an LLM or a knowledge base (UMLS) to link diagnoses, symptoms, risk factors, and treatments. Such a KG will make it possible to reason over relationships between entities.
- **Hybrid TF + GNN model**: combine a powerful sequential model (Transformer or GRU) for temporal data with a GNN over the KG. This takes advantage of the strengths of KindMed and HypeMed. The GNN can integrate both structured EHR data and the KG, while the Transformer captures chronological dependencies.
- **Multi-view architecture**: consider the technique used by MGRN and HypeMed: segment the views (e.g., "current visit," "history," and "code transformation" views) and use a gating retrieval mechanism to draw critical information from similar previous visits.
- **DDI safety**: implement a DDI penalty module (as in SafeDrug) or add negative constraints during training (KEHGCN). You could have an algorithm that filters or adjusts the final recommendations to eliminate high-risk pairs, relying on official sources (e.g., DrugBank).

- **Attribution-based explainability:** integrate methods such as LIME [43, L59-L63] and GNNExplainer [36, L141-L144] to identify which patient attributes and which KG relations influenced the prediction. These quantified attributions will serve as the basis for the explanation.
- **Evidence chain and clinical rules:** complement attribution with formal medical rules (treatment protocols, guidelines) and literature excerpts. For example, if LIME/GNNExplainer identifies "hyperglycemia" and "history of diabetes" as factors, the explanation should recall the clinical rule "mannitol is prescribed in cases of severe hyperglycemia" (provided by a knowledge base). This grounds the explanation in real evidence.
- **Distinction between recommendations and explanations:** follow the PIE paradigm. Separate the recommendation engine (neural) from the explanation engine. Once the candidate medication list has been produced, build an independent explanation based only on attribution scores, KG paths, and clinical sources. An LLM can formulate the explanation, but it must not be the core of the recommendation.
- **Use of LLMs for explanation only:** an LLM (or medical chatbot) can consolidate information and write the final explanation. For example, after retrieving metapaths and attribution scores, one can build a prompt such as: "Explain why A is prescribed based on X, Y, and Z factors." The COMPASS system shows that such a model, fine-tuned on justification examples, can produce relevant and understandable summaries.
- **Comprehensive evaluation:** validate on MIMIC-III/IV and eICU according to best practices (patient split, temporal filter, etc.). Measure both recommendation quality (F1 accuracy, Jaccard against real prescriptions) and explainability metrics: for example, check whether the factors cited in the explanation actually correspond to known reasons (case studies, experts). You could also have clinicians evaluate the relevance of explanations.
- **Error monitoring and iteration:** given the weaknesses of these works, make sure to detect contradictions (for example, if the KG and the literature diverge) and report uncertainty. Integrate a mechanism that enables the system to recognize when it has low confidence (for example, distribution anomalies or limited overlap with the training data) and return warnings to the clinician.

In summary, the essential ideas to remember are: explicitly integrate medical knowledge (graphs/ontologies), model visit temporality and complex interactions (hypergraphs, multi-view modeling), control DDIs, and clearly separate the recommendation and explanation phases. Explanations must be built from genuine model signals (attributions, KG paths, rules) and then formulated naturally (via an LLM), not "invented" by the LLM alone. The richest synthesis will combine the strengths of these approaches: explicit medical knowledge (GraphCare, HiRef), robust modeling (HypeMed, MGRN), safety (SafeDrug, KEHGCN), and controlled explainability (PIE, EGNet).

Key references: The concepts discussed above come mainly from PIE-Med [36, L133-L142] [36, L145-L154], GraphCare [29, L13-L21], KindMed [38, L59-L67], HiRef [4, L61-L70], KEHGCN [6, L17-L25] [6, L27-L30], HypeMed [8, L60-L68], SafeDrug [12, L57-L64], GAMENet [14, L55-L63], MGRN [23, L169-L177], CausalMed [10, L64-L72], EGNet [41, L52-L60], and LIME [43, L59-L63].